

LINJI (JOEY) WANG

Robot Learning Researcher | Robotics Systems Engineer

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PROFILE

Computer Science Ph.D. researcher connecting adaptive RL training with robot policy integration. Research spans curricula over tasks, auxiliary rewards, and domain randomization, with first/co-first-author papers at IROS 2025. Engineering experience includes C++/ROS 2/ONNX inference and C/Python systems at AWS.

TECHNICAL EXPERTISE

Robot learning	Automatic curriculum learning, deep reinforcement learning, PPO, teacher-student learning, reward adaptation, latent representations, domain randomization, sim-to-real
Robotics systems	PyTorch, Isaac Gym, Isaac Lab, MuJoCo, ROS 2, ONNX Runtime; navigation, quadruped locomotion, off-road mobility, humanoid policy inference
Systems & software	C, C++, Python, Aurora PostgreSQL, query processing and join optimization, AWS, Docker, Git/CI-CD

RESEARCH & ENGINEERING EXPERIENCE

RobotiXX Lab, George Mason University

Graduate Research Assistant — Robot Learning

Fairfax, VA

Aug 2023 – Present

- **Task curricula — GACL:** Developed VAE task representations, performance-history conditioning, and grounded task sampling for PPO in 128 parallel Isaac Gym environments. Reported navigation / quadruped simulation success: 81.85% / 79.21% versus CLUTR at 76.67% / 74.65% (first author, IROS 2025).
- **Reward curricula — Reward Training Wheels:** Co-developed adaptive auxiliary-reward weighting. Reported off-road simulation success: 76.67% versus 34.44% with expert rewards; physical completion: 5/5 versus 2/5 trials (co-first author, IROS 2025).
- **Domain-randomization curricula — LUCID (preprint):** Execution-informed humanoid training with a frozen temporal encoder and bounded feedback scheduler. Study reports 88.9% versus 76.8% full-randomization simulation completion against filtered-error PI; G1 at 40 ms added delay: 38/60 versus 23/60 trials.
- **Policy integration:** Implemented residual/base ONNX inference, metadata-driven observations, normalization, and temporal history in an upstream C++/ROS 2 controller for Unitree G1; developing Isaac Lab-MuJoCo parity diagnostics. This integration remains a simulation prototype.
- Co-authored **Moving Through Clutter** (3rd author, preprint 2026), **DDP** (3rd, IROS 2025), and **Adaptive Dynamics Planning** (4th, ICRA 2026). The DDP-based RobotiXX team placed 2nd in both 2025 BARN Challenge phases.

Amazon Web Services (AWS)

Software Development Engineer Intern — Amazon RDS & Aurora

Summers 2025 & 2026

- **Aurora PostgreSQL (2026):** Implemented and extended Adaptive Join in C, adding support for additional join types; built join benchmarks and synthetic-data generators to investigate regressions across engine changes.
- **RDS Proxy (2025):** Built Python/Streamlit performance tooling with multi-region CloudWatch metrics; implemented Welch's tests, power analysis, and Bonferroni correction for regression investigation.

Computational Engineering and Robotics Lab, Carnegie Mellon University

Research Assistant — 3D Perception and AR

Pittsburgh, PA

Jan 2022 – May 2023

- Built an AR scene-inpainting pipeline using GAN image completion plus RANSAC/DBSCAN point-cloud segmentation.

SELECTED PUBLICATIONS

- **Wang, L.**, Xu, Z., Stone, P., Xiao, X. "GACL: Grounded Adaptive Curriculum Learning with Active Task and Performance Monitoring." *IROS 2025*. First author.
- **Wang, L.***, Xu, T.*, Lu, Y., Xiao, X. "Reward Training Wheels: Adaptive Auxiliary Rewards for Robotics Reinforcement Learning." *IROS 2025*. Co-first author.
- **Additional robotics publications:** Moving Through Clutter (3rd author, arXiv 2026); DDP (3rd, IROS 2025); Adaptive Dynamics Planning (4th, ICRA 2026); ColorMap-VIO / II-NVM (5th, RA-L 2026 / 2025).

EDUCATION

George Mason University

Ph.D. in Computer Science — Advisor: Xuesu Xiao

Fairfax, VA

2023 – Expected 2027

Carnegie Mellon University

M.S. in Mechanical Engineering, GPA: 3.94/4.0

Pittsburgh, PA

2021 – 2023

University of Cincinnati

B.S. in Mechanical Engineering, magna cum laude

Cincinnati, OH

2016 – 2021